

26/05/11 Ch15 作業——運算子的多載 operator overloading

Problem 15-8, 11

目錄

8. 「>」 運算子的多載	2
題目敘述	2
題解說明	2
完整程式碼	3
11. 運算子 「=」 的多載	4
題目敘述	4
題解說明	4
完整程式碼	4

8. 「>」運算子的多載

題目敘述

請仿照 Ch15_2，撰寫 operator<() 的三個多載函數。請參考下面的執行結果：

```
1 win1 is larger than win2
2 win1 is smaller than 7000
3 win2 is larger than 4500
```

```
01 // Ch15_2, 運算子「>」多載的範例
02 #include <iostream>
03 using namespace std;
04 class CWin{                                // 定義視窗類別 CWin
05     private:
06         char id;
07         int width, height;
08     public:
09         CWin(char i,int w,int h):id(i),width(w),height(h) // 建構子
10     {}
11     bool operator>(CWin &win){              // 定義運算子「>」的多載
12         return(this->area() > win.area());
13     }
14     int area(void){
15         return width*height;
16     }
17 };
18
19 int main(void){
20     CWin win1('A',70,80);
21     CWin win2('B',60,90);
22
23     if(win1>win2)                            // 判別 win1 與 win2 物件面積的大小
24         cout << "win1 is larger than win2" << endl;
25     else
26         cout << "win2 is larger than win1" << endl;
27     return 0;
28 }
• 執行結果：
win1 is larger than win2
```

題解說明

請翻閱 Ch15 PPT，你會發現 Ch15_3 的程式碼就是這題的做法。

```
01 // Ch15_3, 運算子多載的範例(二)
02 #include <iostream>
03 using namespace std;
04 class CWin{                                // 定義視窗類別 CWin
05     private:
06         char id;
07         int width, height;
08     public:
09         CWin(char i,int w,int h):id(i),width(w),height(h) // 建構子
10     {}
11     int operator>(CWin &win){
12         return(this->area() > win.area());
13     }
14     int operator>(const int &var){
15         return(this->area() > var);
16     }
17     int area(void){
18         return width*height;
19     }
20 };
21
22 int operator>(const int &var, CWin &win){
23     return(var > win.area());
24 }
25
26 int main(void){
27     CWin win1('A',70,80);
28     CWin win2('B',60,90);
29
30     if(win1>win2) // 呼叫第一個 operator>() 函數
31         cout << "win1 is larger than win2" << endl;
32     else
33         cout << "win1 is smaller than win2" << endl;
34
35     if(win1>7000) // 呼叫第二個 operator>() 函數
36         cout << "win1 is larger than 7000" << endl;
37     else
38         cout << "win1 is smaller than 7000" << endl;
39
40     if(4500>win2) // 呼叫第三個 operator>() 函數
41         cout << "win2 is smaller than 4500" << endl;
42     else
43         cout << "win2 is larger than 4500" << endl;
44     return 0;
45 }
• 執行結果：
win1 is larger than win2
win1 is smaller than 7000
win2 is smaller than 4500
```

可改寫成
if(win1.operator>(win2))

第一個 operator>() 函數

第二個 operator>() 函數

可改寫成
if(win1.operator>(7000))

第三個 operator>() 函數，注意
這個函數定義在 CWin 類別
之外，因此它是一般的函數

```
win1 is larger than win2
win1 is smaller than 7000
win2 is larger than 4500

Program finished with exit code 0
Press any key to exit
```

完整程式碼

```
1 #include <iostream>
2 using namespace std;
3
4 class CWin
5 {
6     private:
7         char id;
8         int width, height;
9
10    public:
11        CWin(char i, int w, int h) : id(i), width(w), height(h) {}
12
13        int area()
14        {
15            return width * height;
16        }
17
18        // 1.
19        bool operator<(CWin& win)
20        {
21            return this->area() < win.area();
22        }
23
24        // 2. and 3.
25        bool operator<(const int& x)
26        {
27            return this->area() < x;
28        }
29 };
30
31 int main()
32 {
33     CWin win1('A', 70, 80); // area = 5600
34     CWin win2('B', 60, 90); // area = 5400
35
36     // 1.
37     if (win1 < win2) { cout << "win2 is larger than win1" << endl; }
38     else { cout << "win1 is larger than win2" << endl; }
39
40     // 2.
41     if (win1 < 7000) { cout << "win1 is smaller than 7000" << endl; }
42     else { cout << "win1 is larger than 7000" << endl; }
43
44     // 3.
45     if (win2 < 4500) { cout << "win2 is smaller than 4500" << endl; }
46     else { cout << "win2 is larger than 4500" << endl; }
47
48     return 0;
49 }
```

11. 運算子「=」的多載

題目敘述

請依序完成下列題目的需求。

1. 建立座標類別 `coord`，成員包含座標 `x` 和 `y`，另有一個 `show()` 函數可將座標列印在螢幕上。
2. 撰寫 `coord()` 建構子，預設的座標是 `(0, 0)`，並於 `main()` 宣告 `c0(1, 1)`、`c1(2, 2)`、`c2`、`c3`。
3. 請多載「+」、「-」與「=」運算子，使得 `c2 = c0 + c1`，`c3 = c0 - c1`。

請參考下面的執行結果：

```
1 c0=(1,1)
2 c1=(2,2)
3 c0+c1=(3,3)
4 c0-c1=(-1,-1)
```

題解說明

照題意模擬即可，習慣它的語法糖。

```
c0=(1,1)
c1=(2,2)
c0+c1=(3,3)
c0-c1=(-1,-1)

Program finished with exit code 0
Press any key to exit
```

完整程式碼

- (c) 這些方法都是可行的，不過個人偏好：回傳匿名物件。
- 「`const` 修飾它左邊的東西」；如果左邊沒有東西，才修飾右邊。

```
1 #include <iostream>
2 using namespace std;
3
4 class coord
5 {
6     private:
7         int x;
8         int y;
9
10    public:
11        // (b) 建構子，預設座標為 (0,0)
12        coord(int i = 0, int j = 0) : x(i), y(j) {}
13
14        // (a) 印出座標
15        void show()
16        {
17            cout << "(" << x << "," << y << ")" << endl;
18        }
```

```

19
20 // (c) 多載「+」運算子：傳回匿名物件
21 coord operator+(coord const& b)
22 {
23     return coord(this->x + b.x, this->y + b.y);
24 }
25
26 // 多載「-」運算子：傳回新物件 temp
27 coord operator-(coord const& b)
28 {
29     coord temp;
30     temp.x = this->x - b.x;
31     temp.y = this->y - b.y;
32
33     return temp;
34 }
35
36 // 多載「=」運算子：為了連鎖賦值，必須傳回 *this 的 reference
37 coord& operator=(coord const& b)
38 {
39     this->x = b.x;
40     this->y = b.y;
41
42     return *this; // 傳回自己本人 (reference)
43 }
44 };
45
46 int main()
47 {
48     coord c0(1, 1);
49     coord c1(2, 2);
50     coord c2;
51     coord c3;
52
53     cout << "c0=";    c0.show();
54     cout << "c1=";    c1.show();
55
56     c2 = c0 + c1;
57     cout << "c0+c1=";  c2.show();
58
59     c3 = c0 - c1;
60     cout << "c0-c1=";  c3.show();
61
62     return 0;
63 }

```